ASEEM RAJ BARANWAL

Last updated: July 7, 2025

EDUCATION

Ph.D. in Computer Science, University of Waterloo — 2020 – 2024
Supervised by Aukosh Jagannath and Kimon Fountoulakis

M.Math. (thesis) in Computer Science, University of Waterloo — 2018 – 2020
Supervised by Jeffrey Shallit

B.Tech. in Computer Science, Indian Institute of Technology Jodhpur — 2012 – 2016

EXPERIENCE

- **AI Resident**, XTX Services LLC — Jan 2025 – Present
- **Research Intern**, Microsoft Research — Jun 2023 – Sep 2023
- **Research Intern**, Google Research — Oct 2022 – Dec 2022
- **Software Engineer**, Microsoft Corporation — Jun 2016 – Jul 2018

PUBLICATIONS (Scholar Profile)

- **Analysis of Corrected Graph Convolutions**. R. Wang, A. Baranwal, K. Fountoulakis. Neural Information Processing Systems (NeurIPS), 2024.
- **Optimality of Message-passing Architectures for Sparse Graphs**. A. Baranwal, K. Fountoulakis, A. Jagannath. Neural Information Processing Systems (NeurIPS), 2023.
- **Graph Attention Retrospective**. K. Fountoulakis, A. Levi, S. Yang, A. Baranwal, A. Jagannath. Journal of Machine Learning Research (JMLR) vol. 24:246 pp. 1–52, 2023.
- **Effects of Graph Convolutions in Multi-layer Networks**. A. Baranwal, K. Fountoulakis, A. Jagannath. International Conference on Learning Representations (ICLR), 2023. *(Spotlight)*
- **Graph Convolution for Semi-supervised Classification: Improved Linear Separability and Out-of-distribution Generalization**. A. Baranwal, K. Fountoulakis, A. Jagannath. International Conference on Machine Learning (ICML), 2021. *(Spotlight)*
- **Antisquares and Critical Exponents**. A. Baranwal, J. Currie, L. Mol, P. Ochem, N. Rampersad, J. Shallit. Discrete Mathematics & Theoretical Computer Science vol. 25:2, 2023.
- **Ostrowski-automatic Sequences: Theory and Applications**. A. Baranwal, L. Schaeffer, J. Shallit. Theoretical Computer Science 858, pp. 122–142, 2021.
- **Decision Algorithms for Ostrowski-automatic Sequences.** A. Baranwal. MMath Thesis, University of Waterloo, School of Computer Science, 2020.
- **Repetitions in Infinite Palindrome-rich Words**. A. Baranwal, J. Shallit. In Combinatorics on Words. WORDS 2019. LNCS vol. 11682, Springer, pp. 93–105, 2019.
- **Critical Exponent of Balanced Words via the Pell Number System**. A. Baranwal, J. Shallit. In Combinatorics on Words. LNCS vol. 11682, Springer, pp. 80–92, 2019.

AWARDS AND ACHIEVEMENTS

- Distinguished Dissertation Award, Cheriton School of Computer Science, UWaterloo — 2025
- Cheriton Graduate Scholarship, University of Waterloo (10,000 CAD) — 2024
- President's Graduate Scholarship, University of Waterloo (10,000 CAD) — 2023–2024
- Ontario Graduate Scholarship, Province of Ontario and UWaterloo (15,000 CAD) — 2023–2024
- Top reviewer award, Learning on Graphs conference — 2022, 2024
- Best paper award, GroundedML workshop at ICLR — 2022
- Graduate Excellence Award, University of Waterloo — 2020–2022
- Graduate Research Travel Grant, GSPA, University of Waterloo — 2019
- Microsoft Engineering Star Award – Changing Status Quo — 2017

TALKS

- Locally Optimal Message-passing on Feature-decorated Sparse Graphs, BIRS, Banff — May 2024
- Optimality of Message-passing Architectures for Sparse Graphs, NeurIPS, New Orleans — Dec 2023
- Polymorphic Malware Classification using Graph Neural Networks, Microsoft Research — Aug 2023
- Effects of Graph Convolutions in Multi-layer Networks, ICLR (virtual talk) — Apr 2023
- Graph Convolutions for semi-supervised learning, ICML (virtual talk) — Jul 2021
- Repetitions in infinite palindrome-rich words, Loughborough University, England — Sep 2019
- Anomaly detection and monitoring of metrics in Azure services, Microsoft Corporation — Jan 2017

RELEVANT COURSEWORK

- Introduction to Machine Learning
- Optimization for Data Science
- Theory of Probability
- Spectral Graph Theory and High-Dimensional Expanders
- Advanced Topics in Data Structures
- Advanced Topics in Distributed Computing
- Software Foundations

TEACHING

- Instructional Apprentice, CS 341 - Algorithms — 2019–2024
- Teaching Assistant, CS 135 - Designing Functional Programs — 2018–2019

MISCELLANEOUS

- Served as a reviewer for NeurIPS, ICLR, LoG, UAI — 2021–Now
- Received talent bursaries from the Alberta Machine Intelligence Institute — 2023, 2024
- Helped underprivileged kids with education at SOS Children's Villages of India — 2016–2018
- Problem setter for ACM ICPC preliminary contests in India — 2015–2016